

IT Infrastructure Points

1 Networking Fundamentals

1. Switches manage internal traffic flow between servers inside a data center.
2. Packet switching problems inside server racks often point to switch-related issues.
3. WAN is used to connect offices across different cities or large geographic distances.
4. LAN is ideal for high-speed communication inside a single building or facility.
5. IPv6 provides a much larger address space than IPv4 for future growth.
6. Addresses such as 2001:1ac2:: represent IPv6 addressing format.

2 Compute and Server Hardware

1. The compute layer is responsible for processing power and running applications on servers.
2. Blade servers provide high compute density while using minimal rack space.
3. iDRAC enables remote power control, health checks, and out-of-band server management.
4. iLO allows administrators to monitor hardware status such as temperature, fans, and power remotely.
5. Fan failure alerts should be treated seriously because they can lead to overheating and hardware damage.

3 Storage Infrastructure

1. DAS connects storage directly to a single server for local use cases such as backups.
2. NAS provides centralized shared storage over an Ethernet network.
3. FCoE supports high-speed, low-latency SAN communication by carrying Fibre Channel over Ethernet.
4. RAID 1 provides disk redundancy by mirroring data so it remains available after one disk failure.

4 Operating Systems and Administration

1. Windows 10 is commonly used for client workstations, while Windows Server 2016 is used for server roles.
2. A domain-based environment is better than a workgroup when centralized authentication and permissions are required.
3. NTFS is preferred on Windows Server volumes because it supports large files, permissions, and security features.
4. Datacenter management consoles can be used to deploy operating systems remotely on bare-metal servers.

5 Security and Protection

1. Server protection should include both hardware and software firewall controls.
2. Infrastructure security must cover unauthorized access, malware, and network-based attacks.